

Estimation Theory - Exercise 4

- Submission date: 2/7/26 at 23:59 via the Moodle website.
 - Keep a clear and organized presentation, handwritten or typed.
 - Unexplained results will not be accepted.
 - Please write your name and I.D. or passport number on the first page.
1. The i.i.d. random variables $x_1, \dots, x_N$ are Poisson-distributed with parameter λ , i.e. their probability mass function is $P(x_n = k; \lambda) = e^{-\lambda} \frac{\lambda^k}{k!}$, $n = 1, \dots, N$, $k = 0, 1, 2, \dots$
 - a. Find a minimal sufficient statistic for estimation of λ . Prove that it is minimal.
 - b. Find the maximum likelihood (ML) estimator of λ .
 2. The i.i.d. random variables $x_1, \dots, x_N$ have probability density function (pdf)

$$f(x_n; \mu, \sigma) = \frac{1}{2\sigma} \exp \left\{ -\frac{|x_n - \mu|}{\sigma} \right\}, \quad \mu \in \mathbb{R}, \quad 0 < \sigma < \infty.$$

In Sections a and b, assume the parameter μ is unknown, and σ is known.

- a. Find a sufficient statistic for estimation of μ .
- b. Find the ML estimator of μ .

In Sections c and d, assume the parameter μ is known, and σ is unknown.

- c. Find a sufficient statistic for estimation of σ . Find the ML estimator of σ .
 - d. Find the bias and variance of the ML estimator of σ .
3. The random variables $x_1, \dots, x_N$ are i.i.d. Binomial distributed with parameter θ , i.e. their pdf is $f(x_n; \theta) = \sum_{k=0}^L \binom{L}{k} \theta^k (1 - \theta)^{L-k} \delta(x_n - k)$, where $0 \leq \theta \leq 1$.
 - a. Find the ML estimator of θ from the observations $x_1, \dots, x_N$.
 - b. Is the ML estimator efficient?
 4. Let $\{x_n\}_{n=1}^N$ be a sequence of Bernoulli distributed discrete random variables, i.e. the probability mass function of x_n is given by $p_\alpha(\theta) = \theta^\alpha (1 - \theta)^{1-\alpha}$, $\alpha = 0, 1$, and $0 \leq \theta \leq 1$ is a deterministic parameter. We are interested in the estimation of θ given $\mathbf{x} = [x_1, \dots, x_N]^T$. Define: $t(\mathbf{x}) = \sum_{n=1}^N x_n$.
 - a. Show that $t(\mathbf{x})$ is a sufficient statistic for estimation of θ .
 - b. Is $t(\mathbf{x})$ a minimal sufficient statistic for estimation of θ ? Prove.
 - c. Is $t(\mathbf{x})$ a complete sufficient statistic? Prove.

Hint: $t(\mathbf{x})$ has a binomial distribution, i.e. the probability mass function of $t(\mathbf{x})$ is

$$\Pr\{t = m\} = \binom{N}{m} \theta^m (1 - \theta)^{N-m}.$$

5. The joint pdf of $\mathbf{x} = [x_1, x_2]^T$ is given by

$$f(\mathbf{x}; \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp \left\{ -\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)} \right\}.$$

We are interested to estimate the correlation coefficient ρ using N statistically independent observations of $\mathbf{x}$: $\mathbf{x}_1, \dots, \mathbf{x}_N$, $\mathbf{x}_n = [x_{1,n}, x_{2,n}]^T$.

- Find the likelihood function for estimation of ρ .
- Find a minimal sufficient statistic for estimation of ρ .
- Find a lower bound on the variance of any unbiased estimator of ρ .
- Is the ML estimator of ρ efficient?

6. $x_1, \dots, x_N$ are i.i.d. von Mises distributed random variables with pdf

$$f(x_n; \mu, \kappa) = \begin{cases} \frac{1}{2\pi I_0(\kappa)} \exp \{ \kappa \cos(x_n - \mu) \}, & -\pi \leq x_n < \pi, \\ 0, & \text{else,} \end{cases}$$

where $\mu \in [-\pi, \pi)$, $\kappa > 0$, and $I_0(\kappa) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{\kappa \cos(x)} dx$ is the modified Bessel function of order zero. This distribution is the periodic version of Gaussian distribution.

Given $x_1, \dots, x_N$:

- Find a two-dimensional sufficient statistic for estimation of μ , in case κ is known.
- Find the ML estimator of μ , in case κ is known.
- Find the ML estimator of κ , in case μ is known. It is sufficient to obtain an equation that the ML estimator of κ must satisfy.

7. The random variables $x_1, \dots, x_N$, $N > 1$, are statistically independent uniformly distributed in the interval $[\theta_1, \theta_2]$. We are interested in the estimation of $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$.

- Find the likelihood function for estimation of $\boldsymbol{\theta}$. Find a sufficient statistic for estimation of $\boldsymbol{\theta}$.
- Find the ML estimator of $\boldsymbol{\theta}$.
- Is $\hat{\boldsymbol{\theta}}_{\text{ML}}$ biased? If it is, propose an unbiased estimator based on $\hat{\boldsymbol{\theta}}_{\text{ML}}$.
- Find a minimum variance unbiased MVU estimator of $\boldsymbol{\theta}$. Prove that it is an MVU estimator.

Hint: For $r = \min \{x_n\}_{n=1}^N$, $q = \max \{x_n\}_{n=1}^N$, then

$$F_{r,q}(a, b) = \Pr(r \leq a, q \leq b) = \Pr(q \leq b) - \Pr(r > a, q \leq b).$$